WORKED SOLUTION

STEP II 1992, Question 8

Pure • Techniques of Integration

STEP II 1992, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q8

The key idea

Three standard techniques, one per part. In (i) we factorise the denominator completely and notice a *repeated* factor, so the partial fractions need a squared term. In (ii) the substitution $t = \tan\frac{x}{2}$ turns any rational function of $\sin x$ and $\cos x$ into a rational function of t, and here the resulting quadratic factorises nicely. In (iii) we write $\sinh x$ in exponentials and substitute $u = \mathrm{e}^x$, which again leaves partial fractions.

Solution

Part (i)

Since $x^2 - 1 = (x-1)(x+1)$, the denominator is $(x-1)^2(x+1)$. The factor $x-1$ is repeated, so we look for

$$\frac{x}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1},$$

that is,

$$x = A(x-1)(x+1) + B(x+1) + C(x-1)^2.$$

Putting $x = 1$ gives $1 = 2B$, so $B = \frac{1}{2}$. Putting $x = -1$ gives $-1 = 4C$, so $C = -\frac{1}{4}$. Comparing coefficients of x^2 gives $0 = A + C$, so $A = \frac{1}{4}$. Hence

$$\begin{aligned}\int \frac{x}{(x-1)(x^2-1)}\,\mathrm{d}x &= \int \left(\frac{1}{4(x-1)} + \frac{1}{2(x-1)^2} - \frac{1}{4(x+1)}\right)\mathrm{d}x \\ &= \tfrac{1}{4}\ln|x-1| - \frac{1}{2(x-1)} - \tfrac{1}{4}\ln|x+1| + c \\ &= \frac{1}{4}\ln\left|\frac{x-1}{x+1}\right| - \frac{1}{2(x-1)} + c.\end{aligned}$$

Part (ii)

Let $t = \tan\frac{x}{2}$. Then

$$\cos x = \frac{1-t^2}{1+t^2}, \qquad \sin x = \frac{2t}{1+t^2}, \qquad \mathrm{d}x = \frac{2\,\mathrm{d}t}{1+t^2},$$

the last because $\mathrm{d}t/\mathrm{d}x = \frac{1}{2}\sec^2\frac{x}{2} = \frac{1}{2}(1+t^2)$. So

$$3\cos x + 4\sin x = \frac{3-3t^2+8t}{1+t^2} = \frac{(1+3t)(3-t)}{1+t^2},$$

and the factors $1+t^2$ cancel:

$$\int \frac{\mathrm{d}x}{3\cos x + 4\sin x} = \int \frac{2\,\mathrm{d}t}{(1+3t)(3-t)}.$$

? **Ask yourself** How do we know the quadratic $3+8t-3t^2$ will factorise?

Its discriminant is $64+36 = 100 = 10^2$, a perfect square, so its roots $t = 3$ and $t = -\frac{1}{3}$ are rational. (That is no accident: $3^2+4^2 = 5^2$, which is also why the harmonic form below has a whole-number amplitude.)

Now write $\dfrac{2}{(1+3t)(3-t)} = \dfrac{a}{1+3t} + \dfrac{b}{3-t}$, so $2 = a(3-t) + b(1+3t)$. Putting $t = 3$ gives $b = \frac{1}{5}$; putting $t = -\frac{1}{3}$ gives $\frac{10}{3}a = 2$, so $a = \frac{3}{5}$. Therefore

$$\begin{aligned}\int \frac{\mathrm{d}x}{3\cos x + 4\sin x} &= \int \left(\frac{3}{5(1+3t)} + \frac{1}{5(3-t)}\right)\mathrm{d}t = \tfrac{1}{5}\ln|1+3t| - \tfrac{1}{5}\ln|3-t| + c \\ &= \frac{1}{5}\ln\left|\frac{1+3\tan\frac{x}{2}}{3-\tan\frac{x}{2}}\right| + c.\end{aligned}$$

Alternatively, write $3\cos x + 4\sin x = 5\sin(x+\beta)$, where $\cos\beta = \frac{4}{5}$ and $\sin\beta = \frac{3}{5}$. Then the integral is $\frac{1}{5}\int \operatorname{cosec}(x+\beta)\,\mathrm{d}x = \frac{1}{5}\ln|\tan\frac{1}{2}(x+\beta)| + c$. The two answers agree: with $\tan\frac{\beta}{2} = \frac{1}{3}$ (since $\tan\frac{\beta}{2} = \frac{\sin\beta}{1+\cos\beta} = \frac{3/5}{9/5}$), the addition formula gives

$$\tan\tfrac{1}{2}(x+\beta) = \frac{t+\frac{1}{3}}{1-\frac{1}{3}t} = \frac{1+3t}{3-t}.$$

Part (iii)

Since $\sinh x = \frac{1}{2}(\mathrm{e}^x - \mathrm{e}^{-x})$, multiplying top and bottom by e^x gives

$$\frac{1}{\sinh x} = \frac{2\mathrm{e}^x}{\mathrm{e}^{2x}-1}.$$

Substitute $u = \mathrm{e}^x$, so $\mathrm{d}u = \mathrm{e}^x\,\mathrm{d}x$:

$$\begin{aligned}\int \frac{\mathrm{d}x}{\sinh x} &= \int \frac{2\,\mathrm{d}u}{u^2-1} = \int\left(\frac{1}{u-1} - \frac{1}{u+1}\right)\mathrm{d}u = \ln\left|\frac{u-1}{u+1}\right| + c \\ &= \ln\left|\frac{\mathrm{e}^x-1}{\mathrm{e}^x+1}\right| + c = \ln\left|\tanh\frac{x}{2}\right| + c,\end{aligned}$$

where the last step divides top and bottom by $e^{x/2}$: $\dfrac{e^{x/2}-e^{-x/2}}{e^{x/2}+e^{-x/2}} = \tanh\frac{x}{2}$. Note the neat parallel with $\int \operatorname{cosec} x\,\mathrm{d}x = \ln|\tan\frac{x}{2}| + c$ used in the alternative to (ii).

Checks

Differentiate each answer. In (iii), for example, $\frac{\mathrm{d}}{\mathrm{d}x}\ln\tanh\frac{x}{2} = \frac{\frac{1}{2}\operatorname{sech}^2\frac{x}{2}}{\tanh\frac{x}{2}} = \frac{1}{2\sinh\frac{x}{2}\cosh\frac{x}{2}} = \frac{1}{\sinh x}$. Numerically, Simpson's rule gives $\int_2^5$ of the integrand in (i) as 0.548287, $\int_0^1$ of the integrand in (ii) as 0.234276 and $\int_{1/2}^2 \operatorname{cosech} x\,\mathrm{d}x$ as 1.134488, each matching the antiderivative to at least 12 decimal places.

Common slips

- In (i), missing the repeated factor and trying $\frac{A}{x-1} + \frac{C}{x+1}$ alone, or forgetting that $x^2 - 1$ factorises at all.
- Dropping the modulus signs inside the logarithms: the integrands are defined for negative values of $x - 1$, $1 + 3t$ and $\tanh\frac{x}{2}$ too.
- In (ii), forgetting the factor $\frac{2}{1+t^2}$ from $\mathrm{d}x$.

Beyond STEP

The substitution $t = \tan\frac{x}{2}$ (sometimes called the Weierstrass substitution) shows that *every* rational function of $\sin x$ and $\cos x$ has an elementary antiderivative, because partial fractions handle every rational function of t. Behind it is a parametrisation of the unit circle by rational functions, the same one that generates all Pythagorean triples. Questions of which functions have elementary antiderivatives at all lead to Liouville's theorem and differential algebra; for the analytic foundations of integration, see Abbott, *Understanding Analysis*.

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